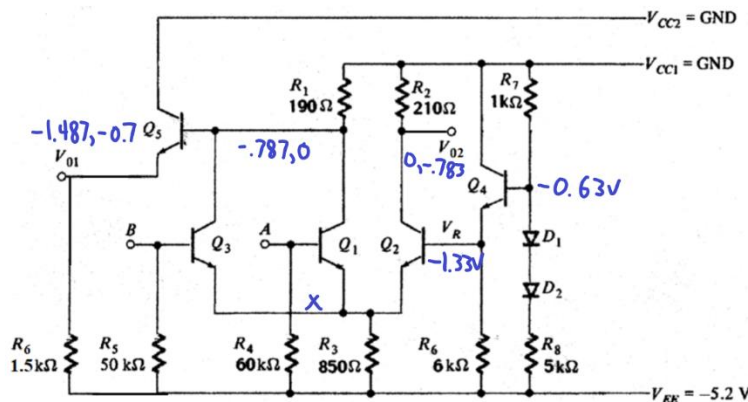


1a)



First find V_R :

$$R_7 + R_8 = 1k + 5k = 6k\Omega \quad V_{R78} = 0 - V_{EE} - V_{D1} - V_{D2} = 0 + 5.2 - 0.7 - 0.7$$

$$I_{R7} = \frac{V_{R78}}{R_{78}} = \frac{3.8}{6k} = 633 \mu A \quad V_{R78} = 3.8V$$

$$V_{B4} = 0 - V_{R7} = -I_{R7} \cdot R_7 = -0.633m \cdot 1k$$

$$V_{B4} = -0.633V$$

$$V_R = V_{B4} - V_{BE} = -0.633 - 0.7$$

$$V_R = -1.33V$$

When A or B is high, $V_x = V_{A,D} - V_{BE}$

$$V_x = -1.33 + 0.35 - 0.7$$

$$V_x = -1.68V$$

$$I_{ss} = \frac{V_x - V_{EE}}{R_7} = \frac{-1.68 - (-5.2)}{850}$$

$$I_{ss} = 4.14mA \text{ when } AB \neq 00$$

$$V_{BAS} = 0V \text{ when } AB = 00$$

$$V_{BAS} = 0 - I_{ss} \cdot R_1 = -4.14m \cdot 190 = -0.787V$$

For V_{01} , $V_{04} = 0 - V_{BE}$ $V_{0L} = -0.787 - V_{BE}$

$$V_{04} = -0.7V \quad V_{0L} = -1.487V$$

When both A and B are low, $V_x = V_R - V_{BE}$

$$V_x = -1.33 - 0.7$$

$$V_x = -2.03V$$

$$I_{ss} = \frac{V_x - V_{EE}}{R_7} = \frac{-2.03 - (-5.2)}{850}$$

$$I_{ss} = 3.73mA \text{ when } AB = 00$$

For V_{02} , $V_{0L} = 0 - I_{ss} \cdot R_2 = 3.73m \cdot 210$

$$V_{0L} = -0.783V$$

$$V_{04} = 0$$

$$V_{IH} \equiv V_R + V_T \ln\left(\frac{V_{SWING}}{V_T} - 2\right)$$

$$V_{IL} \equiv V_R - V_T \ln\left(\frac{V_{SWING}}{V_T} - 2\right)$$

$$V_{SWING} = I_{ssAB=00} \cdot R_1 + I_{ssAB \neq 00} \cdot R_2$$

$$V_{SWING} = 3.73m \cdot 210 + 4.14m \cdot 190$$

$$V_{SWING} = 1.57V$$

$$V_T \ln\left(\frac{V_{SWING}}{V_T} - 2\right) = 0.026 \cdot \ln\left(\frac{1.57}{0.026} - 2\right) = 0.106V$$

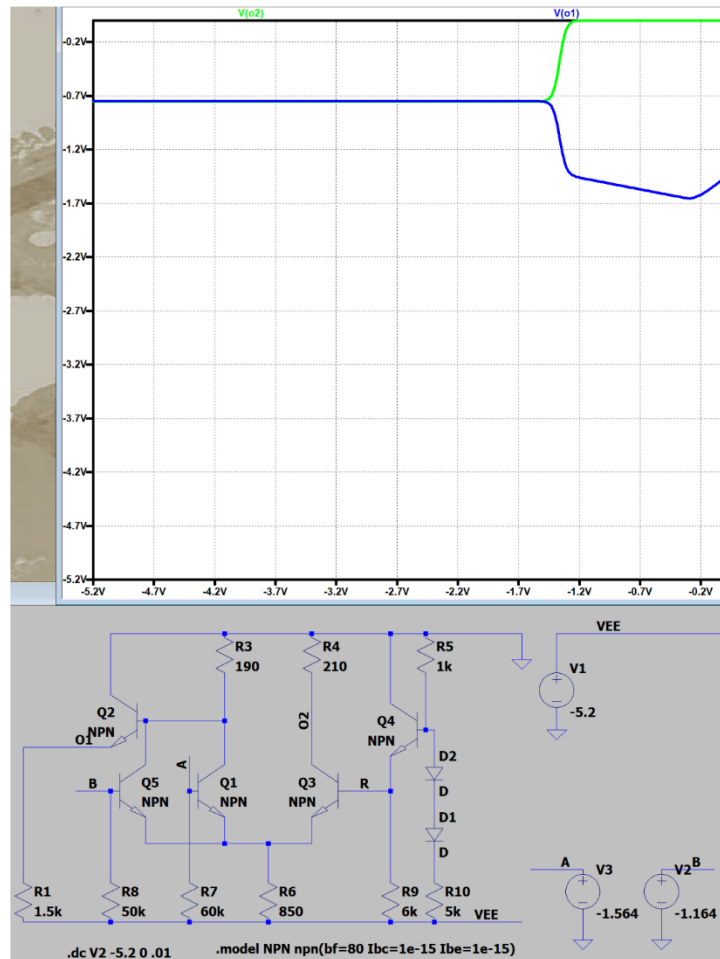
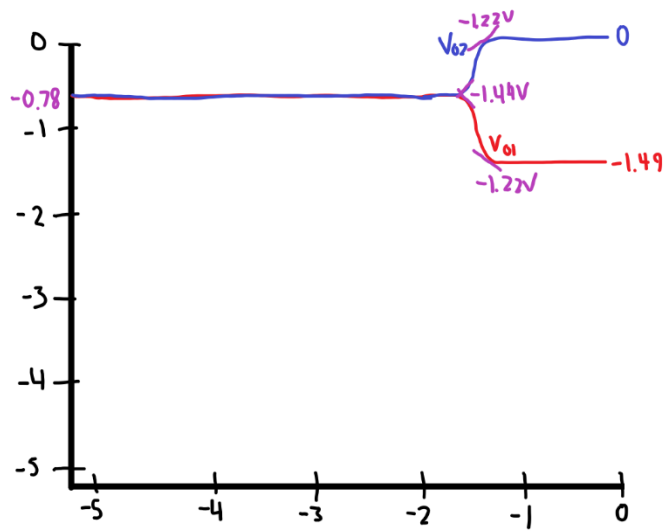
$$V_{IH} = -1.33 + 0.106$$

$$V_{IL} = -1.33 - 0.106$$

$$V_{IH} = -1.224V$$

$$V_{IL} = -1.436V$$

b)



The plots are similar. Simulation shows V_{O1} dropping as the input voltage increases, as expected. Increase in V_{O1} at the tail of the plot indicates that Q3 enters saturation.

c)

$$NM_H = V_{OH} - V_{IH} = -0.7 - -1.224$$

$$NM_H = 0.524V$$

$$NM_L = V_{IL} - V_{OL} = -1.436 - -1.57, NM_L = 0.13V$$

Prior V_{OL} calculation assumed V_{OL} , V_{OH} were centered around V_R , which they are not. V_a with nominal V_{OH} as input yields a V_{OL} of $-1.57V$ according to SPICE plot.

d)

V_{O2} performs an OR operation, V_{O1} performs a NOR operation.

e)

$$NM_H' = \frac{NM_H}{2} = \frac{0.524}{2} = 0.262V$$

$$V_{OH}' = NM_H' + V_{IH} = 0.262 + -1.224V = -0.962V$$

$$N = \frac{I_{OH}}{I_{IH}}$$

With input of V_{OH}' , $V_x = V_{OH}' - V_{EE} = -0.962 - 0.7$

$$V_x = -1.662V$$

$$I_{SS} = \frac{V_x - V_{EE}}{R3} = \frac{-1.662 - -5.2}{850}$$

$$I_{SS} = 4.16mA = I_E$$

$$I_B = \frac{I_E}{\beta + 1} = \frac{4.16mA}{81} = 51.4\mu A$$

$$I_{OH} = I_{EQS} - I_{R6}$$

$$I_{EQS} = (\beta + 1) I_{BAS} = (80 + 1) \frac{0 - (V_{OH}' + V_{BE})}{R1}$$

$$I_{EQS} = 81 \cdot \frac{0 - (-0.962 + 0.7)}{190} = 111.7mA$$

$$I_{R6} = \frac{V_{OH}' - V_{EE}}{R6} = \frac{-0.962 - -5.2}{1.5k} = 2.83mA$$

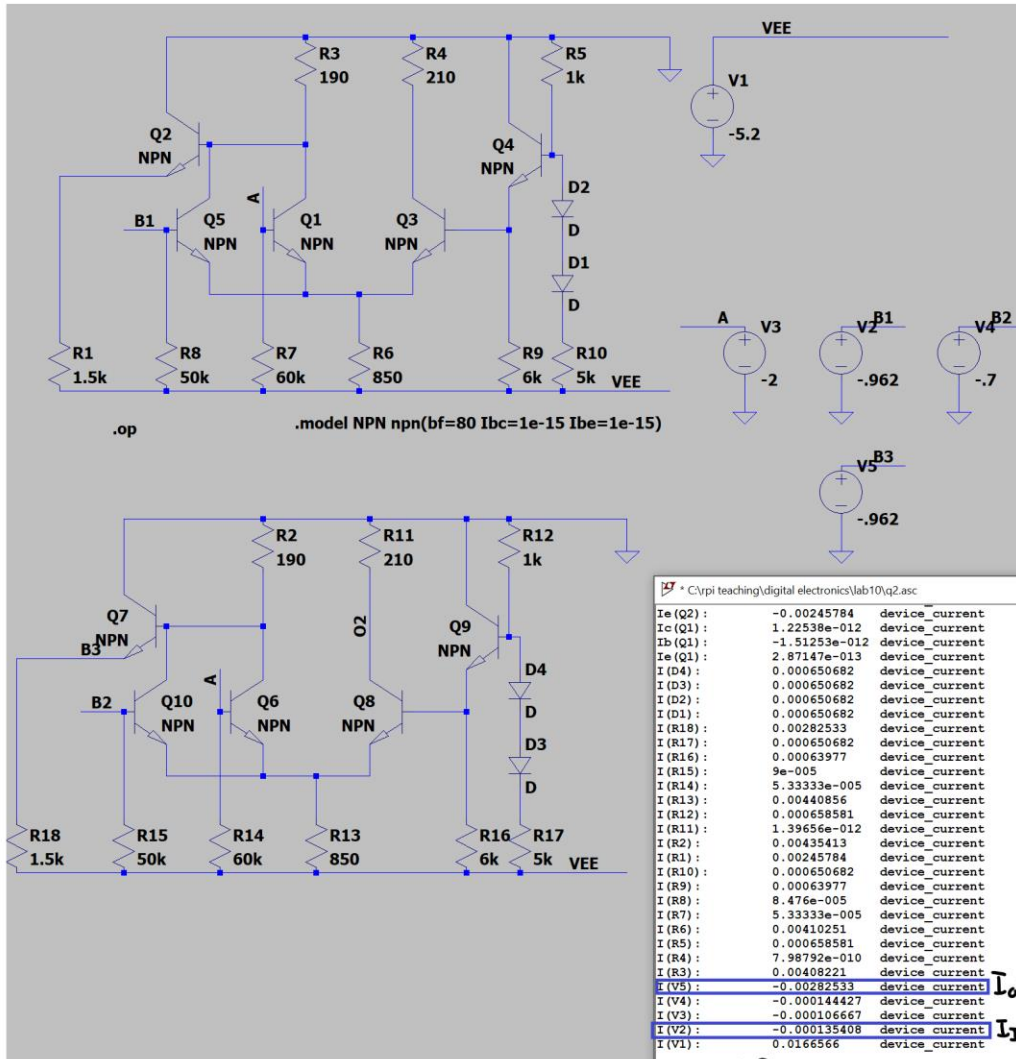
$$I_{OH} = 111.7mA - 2.8mA = 108.9mA$$

At input A,

$$I_{IH} = \frac{V_{OH}' - V_{EE}}{R4} + I_B = \frac{-0.962 - -5.2}{60k} + 51.4\mu A = 122\mu A, N_A = \frac{108.9mA}{122\mu A}, N_A = 893$$

At input B,

$$I_{IH} = \frac{V_{OH}' - V_{EE}}{R5} + I_B = \frac{-0.962 - -5.2}{50k} + 51.4\mu A = 136\mu A, N_B = \frac{108.9mA}{136\mu A}, N_B = 801$$

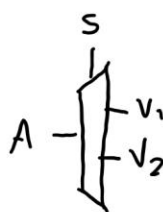


$$N = \frac{2.83 \mu}{0.135 \mu} = 21$$

I_{IH} is as expected, however, I_{OH} and N are much lower. This is because the reverse saturation current defaults in LTSPICE are much lower than required for a V_{BE} of 0.7V with resistances below 1k Ω . I_S needs to be increased to $\sim 1E-13$ to achieve expected hand calculation results.

2)

A	$\overline{v_1}$	$\overline{v_2}$	$\overline{v_1 \overline{v_2}}$
00	1	1	0
01	0	1	1
10	1	1	0
11	0	0	1



This circuit is a demultiplexer/decoder.